



Top 8 CAT Coins and Weights Questions With Video Solutions

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Questions

Instructions

In a game played by two people there were initially N match sticks kept on the table. A move in the game consists of a player removing either one or two matchsticks from the table. The one who takes the last matchstick loses. Players make moves alternately. The player who will make the first move is A. The other player is B.

Question 1

The smallest value of N (greater than 5) that ensures a win for B is

- A 7
- B 6
- C 10
- D 8

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Question 2

The largest of N (less than 50) that ensures a win for B is

- A 46
- B 47
- C 48
- D 49

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Instructions

DIRECTIONS for the following two questions: These questions are based on the situation given below: Ten coins are distributed among four people P, Q, R, S such that one of them gets one coin, another gets two coins, the third gets three coins and the fourth gets four coins. It is known that Q gets more coins than P, and S gets fewer coins than R.

Question 3

If R gets at least two more coins than S, then which one of the following is necessarily true?

- A Q gets at least two more coins than S.
- B Q gets more coins than P.
- C P gets more coins than S.
- D P and Q together get at least five coins.

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Question 4

If the number of coins distributed to Q is twice the number distributed to P then which one of the following is necessarily true?

- A R gets an even number of coins.
- B R gets an odd number of coins.
- C S gets an even number of coins.
- D S gets an odd number of coins.

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Question 5

If Q gets fewer coins than R, then which one of the following is not necessarily true?

- A P and Q together get at least four coins.
- B Q and S together get at least four coins.
- C R and S together get at least five coins.
- D P and R together get at least five coins.

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Instructions

Direction for the following three questions: Answer the questions based on the following information. A, B, C and D collected one-rupee coins following the given pattern. Together they collected 100 coins. Each one of them collected even number of coins. Each one of them collected at least 10 coins. No two of them collected the same number of coins.

Question 6

If A collected 54 coins and B collected two more coins than twice the number of coins collected by C, then the number of coins collected by B could be

- A 28
- B 20
- C 26
- D 22

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Question 7

If A collected 54 coins, then the difference in the number of coins between the one who collected maximum number of coins and the one who collected the second highest number of coins must be at least

- A 12
- B 24
- C 30
- D None of these

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Question 8

The maximum number of coins collected by any one of them cannot exceed

- A 64
- B 36
- C 54
- D None of these

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Answers

1.A

2.D

3.B

4.D

5.A

6.D

7.C

8.A

Explanations

1. **A**

When total matchstick is 1, B will surely win as A is the one who is picking up matchstick first.

When total matchstick is 2, A will surely win as A will pick one matchstick then B has to pick last one.

When total matchstick is 3, A will surely win as he takes up a strategy so that B will have last matchstick to pick up. So A will pick 2 matchstick.

When total matchstick is 4, B will surely win as if A picks up one matchstick, remaining matchsticks will be 3 and we know that when total 3 matchsticks are there then the one, who is picking up matchstick first, is winning surely.

When total matchstick is 5, A will surely win as he takes up a strategy so that B will have last matchstick to pick up. So A will pick 1 matchstick first, then 4 matchsticks will be there and we know that the one, who picks up matchstick second now, will win. Hence A will win.

When total matchstick is 6, A will surely win either he takes up a strategy so that B will have last matchstick to pick up. So A will pick 2 matchstick first, then 4 matchsticks will be there and we know that the one, who picks up matchstick second now, will win. Hence A will win.

Hence it is making a pattern when number of matchsticks are 1,4,7,10.....N then B is winning the game.

So here minimum value of N greater than 5 will be 7.

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2. **D**

When total matchstick is 1, B will surely win as A is the one who is picking up matchstick first.

When total matchstick is 2, A will surely win as A will pick one matchstick then B has to pick last one.

When total matchstick is 3, A will surely win as he takes up a strategy so that B will have last matchstick to pick up. So A will pick 2 matchstick.

When total matchstick is 4, B will surely win as if A picks up one matchstick, remaining matchsticks will be 3 and we know that when total 3 matchsticks are there then the one, who is picking up matchstick first, is winning surely.

When total matchstick is 5, A will surely win as he takes up a strategy so that B will have last matchstick to pick up. So A will pick 1 matchstick first, then 4 matchsticks will be there and we know that the one, who picks up matchstick second now, will win. Hence A will win.

When total matchstick is 6, A will surely win either he takes up a strategy so that B will have last matchstick to pick up. So A will pick 2 matchstick first, then 4 matchsticks will be there and we know that the one, who picks up matchstick second now, will win. Hence A will win.

Hence it is making a pattern when number of matchsticks are 1,4,7,10.....N then B is winning the game.

So largest value of N (less than 50) where B will win, is 49.

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3. **B**

From the given passage, it is given that $Q > P$, $R > S$. Now it is given that R is greater than or equal to $S + 2$. Option B will still remain true because it is given that $Q > P$.

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4. **D**

From the given passage, it is given that $Q > P$, $R > S$.

According to the information $Q = 2$ when $P = 1$ or $Q = 4$ when $P = 2$.

In first case $R = 4$ and $S = 3$ and in the second case $R = 3$ and $S = 1$.

In both the instances S is odd.

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5. **A**

From the given passage, it is given that $Q > P$, $R > S$.

When $R = 4$ and $S = 3$, then $Q = 2$ and $P = 1$.

$P + Q = 2 + 1 = 3$ which contradicts statement 1.

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6. **D**

Number of coins with $A = 54$

Number of coins with $B = 2x + 2$

Number of coins with $C = x$

Number of coins with $D = y$

$$54 + 3x + 2 + y = 100$$

$$3x + y = 44$$

If $2x + 2 = 28$, then $x = 13$ and $y = 5$ which is not possible.

If $2x + 2 = 20$, then $x = 9$ which is not possible.

If $2x + 2 = 26$, then $x = 12$ and $y = 8$ which is not possible

If $2x + 2 = 22$, then $x = 10$ and $y = 14$ which is feasible

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7. **C**

If A collected 54 coins, remaining will be 46.

Now B, C, D must have at least 10, 12, 14 coins

So remaining coins will be 10

So difference between maximum and second maximum should be least when second highest will be as maximum as possible.

$$\text{I.e. difference} = 54 - (14 + 10) = 30$$

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8. **A**

At least number of coin, one can have is 10

And no two of them possess same number of coins, hence for having maximum number of coins to one person, the distribution of coins will be = 10, 12, 14, 64

maximum no. of coins = 64

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